

# GANESH GUPTA

817-822-1131 | iamgs10rk@gmail.com | linkedin.com/in/ganeshhgupta | github.com/ganeshhgupta

## Experience

---

### Oracle

Jun 2025 – Present

*Software Engineer, AI Platform*

- Wrote infrastructure/systems-level code in Go, Rust, and C++ implementing job scheduling, resource allocation, and fault-tolerant placement algorithms for a shared GPU fleet under scarcity constraints.
- Operated multi-cloud fleet orchestration (AWS, GCP, CoreWeave) across distributed, event-driven microservices coordinating training and inference workloads at scale.
- Designed stateful workload recovery via Redis checkpointing; interrupted distributed jobs resume from the last committed step across the fleet.
- Built observability tooling (metrics, logging, event-driven monitoring) tracking fleet utilization, fault detection, and job health, replacing polling-based checks.

### QureAI (AI Startup)

Mar 2026 – Present

*ML Infrastructure Engineer*

- Benchmarked 5+ multi-agent RL architectures on a multi-GPU PyTorch cluster under distributed inference for robustness-critical pipelines.
- Migrated synchronous PyTorch-based model serving to async vLLM with autoscaling, cutting p99 inference latency 420ms to 140ms across 50K+ daily requests.

### Nomura Research Institute Financial Technologies

Sep 2019 – Aug 2023

*Software Engineer*

- Architected and operated 25+ production C++ and SQL-backed microservices in distributed systems for 1M+ users, sustaining 99.9% uptime over 4 years on AWS (EC2, RDS, SQS).
- Drove a redesign of nightly data aggregation, cutting processing from 8 minutes to 90 seconds via indexed views and batch partitioning.
- Standardized failure-handling contracts (retry, timeout, propagation semantics) across 25+ services, reducing production escalations 60% in one quarter.
- Mentored engineers through code reviews and design write-ups, driving consistent engineering standards across the team.

## Projects

---

### Distributed KV Store / VPC Control Plane Simulator | [github.com/ganeshhgupta/distributed-kv-store](https://github.com/ganeshhgupta/distributed-kv-store)

- Raft-consensus key-value store in C++ with leader election, log replication, and log compaction; companion VPC control plane simulator modeling subnet routing for fault-tolerant distributed coordination.

### RootCause AI | [github.com/ganeshhgupta/rootcause-ai](https://github.com/ganeshhgupta/rootcause-ai)

- LangGraph multi-agent system diagnosing production incidents via tool-use chains; routes alerts to specialized sub-agents and synthesizes evidence-backed root cause reports, reducing mean time to resolution.

### Watch-your-LLM | [github.com/ganeshhgupta/watch-your-llm](https://github.com/ganeshhgupta/watch-your-llm)

- Production LLM eval platform with hallucination detection, drift tracking, rubric-based scoring, and regression alerting across model versions; FastAPI backend, PostgreSQL.

## Education

---

### University of Texas at Arlington

Aug 2023 – May 2025

*MS, Computer Science, GPA 4.0/4.0*

### MCKV Institute of Engineering

Jul 2016 – Jul 2020

*BTech, Computer Science*

## Skills

---

**Languages:** Go, C++, Python, Rust, Java

**Scheduling & Distributed Systems:** Job Scheduling, Resource Allocation, Fault-Tolerant Placement, Event-Driven Architecture, Kafka

**Observability & ML:** Metrics/Logging/Tracing, PyTorch, TensorFlow, vLLM

**Cloud & Infra:** AWS, GCP, Docker, Kubernetes, CI/CD

## Certifications

---

AWS Certified Developer Associate | GCP Professional MLOps Engineer